HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 10: From Sensor to Digital Value

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Contents

Reference 6

From Sensor to Digital Value 7

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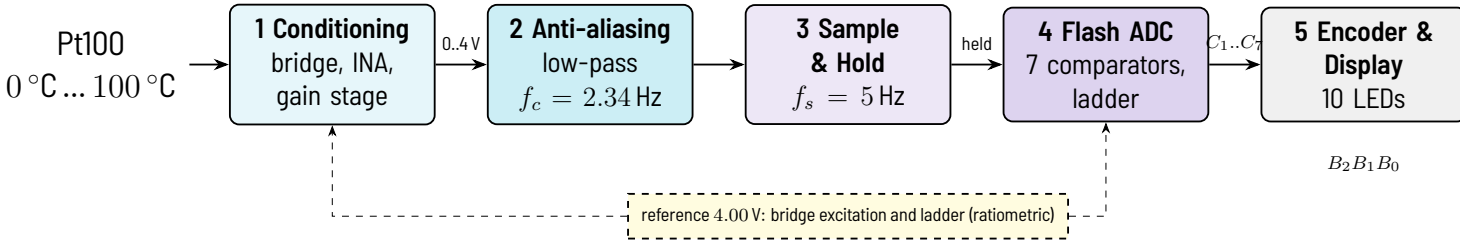
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10 From Sensor to Digital Value

10.1 The Measurement Chain

Five stages from sensor to code



- requirement: $0\,°\text{C} \ldots 100\,°\text{C} \rightarrow 0\,\text{V} \ldots 4.00\,\text{V} \rightarrow$ code 000 …111
- $4\,\text{V}$ full scale makes the step size round: $\text{LSB} = 4\,\text{V}/2^3 = 0.5\,\text{V} \mathrel{\widehat{=}} 12.5\,°\text{C}$
- interference: $50\,\text{Hz}$, $5\,\text{mV}$ in series with the sensor (overhead lamps)

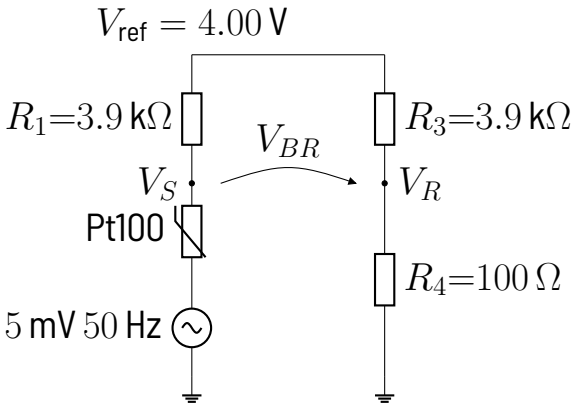
Specification and design values

Quantity	Requirement	Design value
sensor	resistive, slider in Falstad	Pt100, $100\,\Omega$ … $138.5\,\Omega$
sensor current	≤ 1 mA	1.000 mA … 0.990 mA
full scale	0 V …4.00 V	$G = 10 \cdot 10.76 = 107.6$
gain per stage	< 100	1, 10, 10.76
anti-aliasing	$f_c \approx 2.5$ Hz	6.8 kΩ, 10 µF: 2.34 Hz
sampling	$f_s = 5$ Hz	window 20 ms, $C_H = 1$ µF
ADC	3 bit, LSB 0.5 V	8×1 kΩ, 7 comparators
display	LED + resistor	$10\times$ (LED + $330\,\Omega$), 9.8 mA

▶ Falstad: `examples/capstone/capstone.txt`

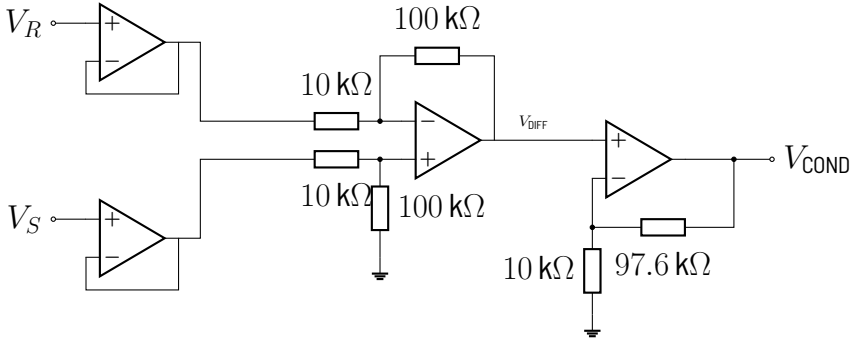
10.2 Stage 1: Signal Conditioning

Bridge design



- sensor current $I_S = V_{\text{ref}}/(R_1 + R) = 4\,\text{V}/4000\,\Omega = 1.000\,\text{mA}$ at $0\,°\text{C}$ ($0.990\,\text{mA}$ at $100\,°\text{C}$)
- balanced at $0\,°\text{C}$: $R_4 = R(0\,°\text{C})$, so $V_{BR} = 0$
- $V_{BR}(T) = V_{\text{ref}}\left(\frac{R(T)}{R_1+R(T)} - \frac{R_4}{R_3+R_4}\right)$, $V_{BR}(100\,°\text{C}) = 37.18\,\text{mV}$

Instrumentation amplifier and gain stage



- buffers: input current zero, the bridge is not loaded
- difference amplifier: $V_{\text{DIFF}} = \frac{100\,\text{k}\Omega}{10\,\text{k}\Omega}(V_S - V_R) = 10\,V_{BR}$
- gain stage: $G_2 = 1 + \frac{97.6\,\text{k}\Omega}{10\,\text{k}\Omega} = 10.76$

Gain: how much, and in how many stages?

- required total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: $G_1 = 10$ (difference amplifier), $G_2 = 4.00/(10 \cdot 0.03718) = 10.758$
  - E96: $R_F = 97.6\,\text{k}\Omega \Rightarrow G_2 = 10.76$, full scale $4.0005\,\text{V}$
  - in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer, adjusted at $100\,°\text{C}$
- finite open-loop gain $A = 10^5$: relative gain error $\approx G/A$
  - one stage $G = 107.6$: $0.11\,\%$; two stages: $2 \times 0.01\,\%$
  - a single stage with $G > 1000$ would lose more than $1\,\%$: keep $G < 100$ per stage

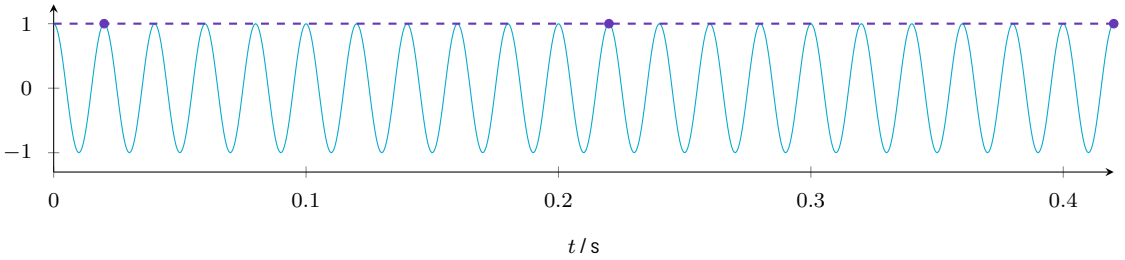
Conditioning

$$V_{\text{COND}} = G_1 G_2\, V_{\text{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right), \quad G_1 G_2 = 107.6$$

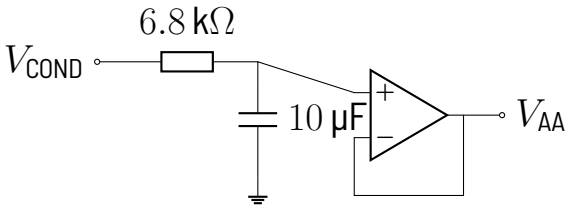
10.3 Stage 2: Anti-Aliasing

Where does the 50 Hz go?

- sampling with f_s folds a frequency f to $f_{\text{alias}} = |f - k f_s|$ (nearest k)
- $f = 50\,\text{Hz}$, $f_s = 5\,\text{Hz}$: $|50 - 10 \cdot 5| = 0\,\text{Hz}$
  - every sample sees the interference at the same phase
  - the result is a **constant offset**: nothing on the scope looks wrong
- amplitude at V_{COND}: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} \cdot 107.6 = 0.52\,\text{V}$ (about 1 LSB)



Anti-aliasing low-pass



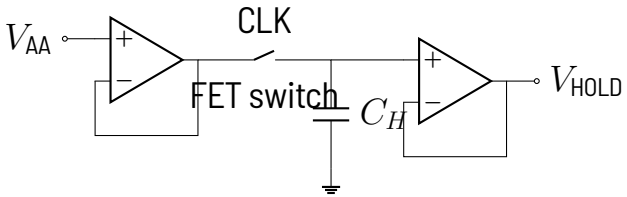
- $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}} = 2.34\,\text{Hz} \leq f_s/2$
- at 50 Hz: $|H| = 1/\sqrt{1 + (50/2.34)^2} = 0.047$ (-26.6 dB)
- residual offset: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250\,\text{mV}$
- the follower decouples the RC network from the S&H input

Design card: Nyquist & anti-aliasing (capstone values)

$f_s = 5$ Hz $\Rightarrow$ bandwidth ≤ 2.5 Hz; $f_{\text{alias}} = |f - k f_s|$; $f_c = 1/(2\pi RC) \leq f_s/2$; residual $= \hat{v}\,/\sqrt{1 + (f/f_c)^2}$ must stay $<$ LSB$/2$.

10.4 Stage 3: Sample and Hold

Buffered sample and hold



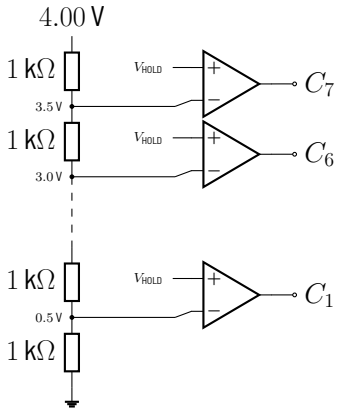
- clock 5 Hz, duty $10\,\%$: switch closed for 20 ms (track), open for 180 ms (hold)
- input buffer: charges C_H quickly; output buffer: no load on C_H

Choosing the hold capacitor

- **charging:** $\tau = R_{\text{on}} C_H = 20\,\Omega \cdot 1\,\mu\text{F} = 20\,\mu\text{s}$
  - settle to LSB$/2$ of 3 bits: $t = \tau \ln 2^4 = 2.8\,\tau = 55\,\mu\text{s} \ll 20\,\text{ms}$
- **droop:** leakage $I_L = 1\,\text{nA}$ (switch + buffer input)
  - $\Delta V = I_L\, t_{\text{hold}} / C_H = 1\,\text{nA} \cdot 0.18\,\text{s} / 1\,\mu\text{F} = 0.18\,\text{mV}$
- both limits leave a wide range: $10\,\text{nF}$ ...$100\,\mu\text{F}$ would work
  - choose a film capacitor with low leakage and dielectric absorption: $C_H = 1\,\mu\text{F}$

10.5 Stages 4 and 5: Flash ADC, Encoder and Display

Reference ladder and comparators



- ladder current $4\,\mathrm{V}/8\,\mathrm{k\Omega} = 0.5\,\mathrm{mA}$; comparator inputs draw no current
- $C_k = 1$ if $V_{\mathrm{HOLD}} > k \cdot 0.5\,\mathrm{V}$; the number of ones is the level 0…7

Quantization and encoder

V_{HOLD} / V	level	C_7	C_6	C_5	C_4	C_3	C_2	C_1	B_2	B_1	B_0
0.0–0.5	0	0	0	0	0	0	0	0	0	0	0
0.5–1.0	1	0	0	0	0	0	0	1	0	0	1
1.0–1.5	2	0	0	0	0	0	1	1	0	1	0
1.5–2.0	3	0	0	0	0	1	1	1	0	1	1
2.0–2.5	4	0	0	0	1	1	1	1	1	0	0
2.5–3.0	5	0	0	1	1	1	1	1	1	0	1
3.0–3.5	6	0	1	1	1	1	1	1	1	1	0
3.5–4.0	7	1	1	1	1	1	1	1	1	1	1

- $B_2 = C_4$ (level ≥ 4)
- $B_1 = C_2\overline{C_4} + C_6$ (levels 2, 3, 6, 7)
- $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$ (odd levels: the edge of the thermometer is at an odd position)

Display

- 7 red LEDs from $C_1 \ldots C_7$: bar graph, a thermometer you can read without decoding
- 3 green LEDs from $B_2 B_1 B_0$: the binary result
- each LED with its own series resistor (chapter 1), driven by a $5\,\mathrm{V}$ logic output: $R = \frac{5\,\mathrm{V} - V_F}{I_F} = \frac{5\,\mathrm{V} - 1.8\,\mathrm{V}}{10\,\mathrm{mA}} = 320\,\Omega \rightarrow$ E12: $330\,\Omega$
  - Falstad: $I_F = 9.8\,\mathrm{mA}$, $V_F = 1.78\,\mathrm{V}$
- never one common resistor for several LEDs: the current would depend on how many are on

10.6 The Complete Chain in Falstad

Using the model

- open ▶ Falstad: `examples/capstone/capstone.txt`
- move the slider from $100\,\Omega$ to $138.5\,\Omega$: the bar graph fills up, the binary LEDs count
- hover over a label to read its voltage; the held value changes every 200 ms
- after a slider step, wait about 0.5 s: the low-pass needs $5\tau = 5 \cdot 68$ ms

Verification table

T / °C	R / Ω	V_{COND} target	V_{HOLD} Falstad	code expected	code Falstad
6.2	102.38	0.25 V	0.251 V	000	000
18.6	107.16	0.75 V	0.751 V	001	001
43.5	116.75	1.75 V	1.749 V	011	011
56.0	121.56	2.25 V	2.250 V	100	100
81.1	131.22	3.25 V	3.249 V	110	110
93.7	136.07	3.75 V	3.750 V	111	111

- the temperatures differ slightly from the linear mapping $25\,°\mathrm{C/V}$ (6, 19, 44, 56, 81, $94\,°$C) because of the bridge nonlinearity

The aliasing experiment

- ▶ Falstad: `examples/capstone/capstone_nofilter.txt`: the S&H takes V_{COND} directly
- V_{COND} carries 0.52 V of 50 Hz; the S&H always samples at the same phase

	V_{COND} (DC)	V_{HOLD}	code
without filter	1.75 V	2.27 V	100 (wrong)
without filter	0.25 V	0.77 V	001 (wrong)
with filter	1.75 V	1.75 V	011

- without the filter the held value is a clean staircase, just 0.52 V too high: one full code too high
- the size of the offset depends on the phase between mains and sampling clock (here the worst case)

10.7 Exercises

Exercise 10.1: From sensor to digital value (capstone task 2025/26)

Build a complete data-acquisition chain in Falstad that turns a slowly changing sensor signal ($0\,°\text{C} \ldots 100\,°\text{C}$) into a 3-bit code. Label the five stages Conditioning | Anti-aliasing | Sample & Hold | Flash ADC | Encoder & Display.

a) Conditioning: resistive sensor in a Wheatstone bridge, sensor current ≤ 1 mA, sensor as a resistor with a slider; instrumentation-amplifier front end (two unity-gain buffers and a difference amplifier); gain trimmed to 0 V at the bottom and exactly 4.00 V at the top of the range; less than 100 per stage.

b) Anti-aliasing: sampling clock $f_s = 5$ Hz. Add the 50 Hz, 5 mV noise source in series with the sensor. Observe the held value without a filter, then add an active low-pass with $f_c \approx 2.5$ Hz and justify f_c from the Nyquist criterion.

c) Sample and hold at 5 Hz: follower, FET switch, hold capacitor, follower. Choose the hold capacitor.

d) Flash ADC: ladder 8×1 kΩ from 4 V, seven comparators, thermometer code.

e) Encoder ($B_2 = C_4$, $B_1 = C_2\overline{C_4} + C_6$, $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$) and display: 7-LED bar graph and three binary LEDs, each LED with its own resistor.

f) Verification table for six inputs near the centres of different code bins, and one line on the aliasing experiment.

Solution 10.1

The design of this chapter is one possible solution; the complete model is ▶ Falstad: `examples/capstone/capstone.txt`.

a) Pt100 with $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$ at $V_{\text{ref}} = 4.00\,\text{V}$: $I_S = 1.000\,\text{mA}$, bridge output $0\,\text{mV} \ldots 37.18\,\text{mV}$. Buffers, difference amplifier $G_1 = 10$, non-inverting stage $G_2 = 1 + 97.6\,\text{k}\Omega/10\,\text{k}\Omega = 10.76$: $V_{\text{COND}} = 0\,\text{V} \ldots 4.00\,\text{V}$.

b) Without filter the 50 Hz aliases to $|50 - 10 \cdot 5| = 0\,\text{Hz}$; the held value is offset by up to $0.52\,\text{V}$ (Falstad: $1.75\,\text{V}$ becomes $2.27\,\text{V}$, code 100 instead of 011, ▶ Falstad: `examples/capstone/capstone_nofilter.txt`). Filter $R = 6.8\,\text{k}\Omega$, $C = 10\,\mu\text{F}$, $f_c = 2.34\,\text{Hz} \leq f_s/2$, 50 Hz attenuated to $4.7\,\%$: $25\,\text{mV}$ residual, below LSB$/2$.

c) $C_H = 1\,\mu\text{F}$: $\tau = 20\,\mu\text{s}$ charges within the $20\,\text{ms}$ window; droop $0.18\,\text{mV}$ per hold period for $1\,\text{nA}$ leakage.

d) Taps at $0.5, 1.0, \ldots, 3.5\,\text{V}$; ladder current $0.5\,\text{mA}$.

e) 3 inverters, 4 AND gates, 2 OR gates (one with 4 inputs); 10 LEDs with $330\,\Omega$, $9.8\,\text{mA}$ each.

f) See the verification table above. Aliasing line: “Without the anti-aliasing filter the held value showed a constant offset of $0.52\,\text{V}$ (one code too high); with the $2.34\,\text{Hz}$ low-pass the offset fell below $30\,\text{mV}$ and all codes were correct.”

Exercise 10.2: Ratiometric measurement

The 4.00 V reference of the capstone is in fact 4.10 V.

a) How large is V_{COND} at $100\,^{\circ}$C?

b) Which code does the converter show at $100\,^{\circ}$C and at $56\,^{\circ}$C? Compare with a design in which the bridge is supplied from a separate, exact 4.00 V source.

Solution 10.2

a) The bridge output scales with the reference: $V_{\mathrm{COND}} = 4.0005\,\mathrm{V} \cdot 4.10/4.00 = 4.10\,\mathrm{V}$.

b) The thresholds scale as well ($k \cdot 0.5125\,\mathrm{V}$). The ratio $V_{\mathrm{COND}}/V_{\mathrm{ref}}$ is unchanged, so both codes are unchanged: 111 at $100\,^{\circ}\mathrm{C}$, 100 at $56\,^{\circ}\mathrm{C}$ ($V_{\mathrm{COND}} = 2.25\,\mathrm{V} \cdot 1.025 = 2.306\,\mathrm{V}$, threshold of level 4: $2.05\,\mathrm{V}$, of level 5: $2.5625\,\mathrm{V}$). With an exact bridge supply and a $4.10\,\mathrm{V}$ ladder the thresholds would be $2.5\,\%$ too high while the signal is correct; a value just above a threshold (e.g. $2.02\,\mathrm{V}$) would then be read one code too low. Ratiometric operation removes the reference from the error budget.

Exercise 10.3: Another sampling rate

The sampling clock of the capstone is changed to $f_s = 4$ Hz and the anti-aliasing filter is removed.

a) To which frequency does the 50 Hz interference alias?
b) Describe what the LEDs show for a constant temperature with $V_{\text{COND}} = 1.75$ V.
c) Is this better or worse than the 0 Hz alias at 5 Hz?

Solution 10.3

a) $|50 - 12 \cdot 4| = 2$ Hz, exactly the Nyquist frequency $f_s/2$: consecutive samples see the interference with a phase step of $2\pi \cdot 50/4 = 25\pi$, so its sign alternates from sample to sample.

b) The held value alternates between $1.75 + \hat{v} \sin\varphi$ and $1.75 - \hat{v} \sin\varphi$ ($\hat{v} = 0.52$ V): the LEDs blink at 2 Hz between two codes, e.g. 100 and 010 in the worst case, or stay at 011 if the phase happens to be $\varphi \approx 0$.

c) It is visible, so it is "better" than a silent offset, but the reading is still wrong. Only the low-pass in front of the sampler removes the error for every sampling rate.

Exercise 10.4: Four bits

The capstone shall be extended to 4 bits with the same 4.00 V full scale.

a) What is the new LSB, in volts and in °C? How many comparators and ladder resistors are needed?
b) Is the first-order anti-aliasing filter still sufficient?
c) Is the bridge nonlinearity still acceptable?

Solution 10.4

a) LSB $= 4\,\mathrm{V}/16 = 0.25\,\mathrm{V} \mathrel{\widehat{=}} 6.25\,^{\circ}\mathrm{C}$; $2^4 - 1 = 15$ comparators, 16 resistors (taps at $k \cdot 0.25\,\mathrm{V}$).

b) Residual $25\,\mathrm{mV} <$ LSB$/2 = 125\,\mathrm{mV}$: yes, still a factor five of margin.

c) The maximum deviation from the straight line is about $0.09\,\mathrm{mV} \cdot 107.6 = 10\,\mathrm{mV} \ll 125\,\mathrm{mV}$: yes.

Exercise 10.5: Hold capacitor at the limits

In the capstone S&H the analog switch has $R_{\text{on}} = 1\,\text{k}\Omega$ and the total leakage current is $10\,\text{nA}$. Determine the smallest and the largest usable hold capacitor if the droop during one hold period and the remaining charging error at the end of the window shall both stay below $\text{LSB}/4 = 125\,\text{mV}$ for a full-scale step.

Solution 10.5

Droop: $I_L\, t_{\text{hold}}/C_H < 0.125\,\text{V} \Rightarrow C_H > 10\,\text{nA} \cdot 0.18\,\text{s}/0.125\,\text{V} = 14.4\,\text{nF}$. Charging: $4\,\text{V} \cdot e^{-t_w/\tau} < 0.125\,\text{V} \Rightarrow t_w > \tau \ln 32 = 3.47\,\tau$, so $\tau < 20\,\text{ms}/3.47 = 5.8\,\text{ms}$ and $C_H < 5.8\,\text{ms}/1\,\text{k}\Omega = 5.8\,\mu\text{F}$. Any value between about $15\,\text{nF}$ and $5.8\,\mu\text{F}$ works; $1\,\mu\text{F}$ is a good choice in the middle (on a logarithmic scale). In Falstad set the switch's on resistance to $1\,\text{k}\Omega$ and C_H to $10\,\mu\text{F}$ in

▶ Falstad: `examples/capstone/capstone.txt`: after a slider step the held staircase needs several samples to reach the new value.